Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

Haben Michael[1]

[1]University of Massachusetts, Amherst, MA (hmichael@math.umass.edu)

Abstract. ... Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1 Introduction

There is currently underway a move to favor generalized linear mixed models in carrying out meta-analyses of event outcomes, particularly in when sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions, such as gaussianity of the effects and perfect knowledge of standard deviations, required by the traditional alternative, inverse-variance weighting. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are suspect, though smaller meta-analyses are where the use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modelling assumptions to improve the efficiency of the testing procedure. When the modelling assumptions are met, the improvement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$. [[the key modelling assumption in this setting is the gaussianity of the random effect.]]

While Bartlett corrections have long been applied to generalized linear models [[cite]], there do not appear to be extensions to even the simple GLMMs that are used in meta-analysis. The reason for the reluctance appears to be computational difficulty [[cite hanada]]. While computer algebra systems have long been applied to compute out the third- and fourth-order derivative arrays needed for Bartlett correction of GLMs ((cite old cordeiro paper, andrews and stafford, mccullagh maybe)), closed-form expressions for GLMM Bartlett corrections are unavailable for any of the commonly used models other than the normal-normal/identity link ((cite noma, michael2026)). A combination of automatic symbolic manipulation and numerical approximation appears necessary. We argue that with careful design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we summarize 4 event data meta-analysis models, following [[Stijnen's]] exposition. In Section ((ref)) we describe a practical approach to estimation of the

Bartlett correction applicfable to these 4 models. In Section ((ref)) we examine the effectiveness and robustness of the corrected test, finding ... . In Section ((ref)) we re-visit the data analyses of ((stijnen)), finding ... .

2 Background

We consider the random effects model for a meta-analysis consisting of n studies. In this model, a Gaussian random variable models the effects $\theta_1, \ldots, \theta_n$, of the individual studies in the meta-analysis,

$$\theta_i = \mu_0 + \tau_0 Z, \qquad Z \sim N(0,1)$$

The target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. The conditional distribution of the data given the random effect depends on the type of analysis being conducted. Following ((stijnen)) we consider 4 types of analyses, determined by whether the studies included are single-arm or two-arm trials, and whether the estimand of the meta-analysis is a proportion or incidence rate.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$f_{m_i}(y_i \mid Z) = \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i}$$
$$p = p(\mu_0, \tau_0^2) = \text{expit}(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as a parameter of the conditional PMF f_{m_i}, which therefore typically varies across the studies in the meta-analysis.[[replace f with fi in displays, update next 3 similarly]]

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The statistical model is conditional on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, under which $Y_i^{(T)}$ follows Fisher's non-central hypergeometric distribution

$$f_{m_i^{(T)}, m_i^{(C)}, S_i}(y_i^{(T)} \mid Z_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}}\omega^{y_i^{(T)}}}{\sum_{\max(0, S_i - m_i^{(C)})}^{\min(S_i, m_i^{(T)})} \binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i - y}\omega^{y}}$$
$$\omega = \omega(\mu_0, \tau_0^2) = \exp(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the ratio of the treatment to control log-odds. The parameters of the conditional PMF are $m_i^{(T)}, m_i^{(C)}$, and S_i.[[ref to bivriate model which models baseline risk, mentions assumptions in this approach]]

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f_{T_i}(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!}$$
$$\lambda = \lambda(\mu_0, \tau_0^2) = T_i \exp(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the log of the incidence rate. The follow-up time T_i is treated as a parameter of the conditional PMF.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. Conditionally on $S_i = Y_i^{(T)} + Y_i^{(C)}$, $Y_i^{(T)}$ is binomial

$$f_{T_i^{(T)},T_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \binom{S_i}{y_i^{(T)}} p^{y_i^{(T)}}(1-p)^{S_i - y_i^{(T)}}$$
$$p = p(\mu_0, \tau_0^2) = \text{expit}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

   The target of inference is the logarithm of the ratio of the incidence rates between the two arms. The sum of events S_i and and the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as parameters of the conditional PMF.

In all cases the likelihood for study i has the form:

$$L_i(\mu, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f_i(Y_i|z)\phi(z)dz, \qquad i = 1, \ldots n$$

where $f_i(y|z)$ is study i's PMF conditional on a Gaussian random effect, as given above. Furthermore, the model parameters (μ, τ^2) only enter the likelihood ((ref)) through a parameter of the conditional PMF, the mean λ in the case of the Poisson, the success probability p in the cases of the binomial, or the odds ratio ω in the case of the non-central hypergeometric.[[maybe put this up]]

The likelihood ratio test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2\sum_{i=1}^{n}(\log L_i(\hat{\mu}, \hat{\tau}^2|Y_i) - \log L_i(\mu_0, \hat{\tau}_0^2|Y_i))$$

where $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto \sum_{i=1}^n \log L_i(\mu, \tau^2|Y_i)$ and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu)$ maximizes $\tau^2 \mapsto \sum_{i=1}^n \log L_i(\mu_0, \tau^2|Y_i)$. In routine applications W converges to a chi-squared distribution under the null hypothesis, by Wilks' Theorem. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.

Under a range of conditions [[citations]] the mean of a correctly specified LR statistic ((ref)) admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

Therefore, the Bartlett-corrected statistic

$$W_B = \frac{W}{1 + b/n}$$

has a mean agreeing with that of a chi-squared random variable up to order $O(1/n^2)$. The work of ((lawley and bartlett)) shows further that, under appropriate conditions all other moments, and so also the entire distribution, agree with a chi-squared up to order $O(1/n^2)$. The resulting asymptotic rate of convergence to a chi-squared random variable is $O(1/n^2)$. As with ((ref lr statistic)), the adjusted statistic may be inverted to obtain CIs and p-values. [[mention C depends on parameters, or maybe make explicit in the notation]]

Among the assumptions underlying the above development we focus on two prominent ones. The first is correct specification of the likelihood, in particular, gaussianity of the unobserved random effect. [[we refer to sitjnen to other modeling assumptions implicit in ((ref 1–4 above)), such as ... .]] While the LR test itself depends on gaussianity for valid inference ((citation showing effects of bad estimatio of tau2)), the Bartlett correction doubles down on the assumption. Second, the arguments under which the Bartlett correction delivers a $1/n^2$ rate are usually justified by Cramér's condition for the underlying random variables. This condition excludes lattice variables, such as are considered here. Consequently, while Bartlett corrections have been derived for statistics based on lattice random variables since at least the 80s, their effectiveness does not rest on a firm theoretical foundation and in fact does not appear to be understood in much generality ((cite frydenberg re multinomial ngative result and gotze, and janson re positive result)). We have no theoretical contibutions to make here and will examine both model robustness and the effect of discreteness by simulation.

[[add a third issue. say at outset 2 issues are really issues with the lr test generally, and the 3rd is about the bartlett correction. wilks theorem applies to interior points. when tau2=0 ie fixed effects model, not applicable. some simulations have suggested does not affect inference for mu, but difficult to see this can be so in general, since inference for tau2 is known to be off ((self,liang)) at the boundary and mu and tau2 are not orthogonal in our models. means for lr test must check tau2 interior is reasonable eg examining the mle, or perhaps computing a fixed effects compute the fixed effects LR CI. [[with iv method can think of re model as a generalization of the fe, if the data is actually fixed effects CIs are valid there is just a loss of power. in the case of the lr, the random effects procedure is different from the fixed effects procedure.]]

3 Method

lawley ((cite)) provides a formula for the correction ((ref)) as sums and products of the components of the inverse Fisher information for ((ref likelihood)) and certain tensors [[k terms]]. Being a small set of sums and products this formula does not introduce numerical or computational difficulties that aren't already present in the basic tensor components, and we omit it [[is not of direct interest here]]. Let D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. For example, $D_{\mu,\tau^2} \sum_{i=1}^{n} \log L_i(\mu, \tau^2 \mid Y_i)$ denotes the off-diagonal term of the obseerved information matrix for ((ref total likelihood)). Define $k_r = \mathbb{E} D_r l(\mu, \tau^2)$ and $k_r^{(s)} = D_s k_r$. In this notation the Fisher information matrix has

components $k_{\mu,\mu}, k_{\mu,\tau^2}, k_{\tau^2,\tau^2}$. Lawley's formula for the Bartlett correction requires all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s of length 1 or r is of length 2 and s of length 1 or 2. Taking into account permutation symmetry of the order of differentiation, there are 35 such terms in a model with one parameter of interest and one nuisance paramter.

Since the meta-analysis data is typically non-identically distributed, the tensor components ((k)) must be computed separately for each study.We consider a single study and drop the subscript i from the conditional PMFs and tensor components ((k)) for ease of exposition.[[also above k referrd to the sum. need to say we are using linearity to compute k's for each study]]. To compute the tensor components ((ref)) for the likelihood ((ref)), we approximate the integral by a quadrature rule. Since all the likelihoods ((ref)) involve integrating against a Gaussian, Gauss-Hermite quadrature is a natural choice, and for the sake of simplicity we adopt here a non-adaptive variant. This and similar methods consist in weights $w_1, \ldots, w_{n_{GH}}$ and nodes in the domain of integration $z_1, \ldots, z_{n_{GH}}$ allowing the integral of smooth integrands like ((ref)) to be approximated by a weighted sum,

$$L(\theta, \tau^2|Y) = \int_{-\infty}^{\infty} f(Y|z)\phi(z)dz \approx \sum_j w_j f(Y|z_j)$$

The number of nodes n_{GH} controls the quality of the approximation and introduces a tuning parameter into the proposed method. A higher-order partial derivative of the log-likelihood may be approximated by $D_r \log L(\mu, \tau^2|Y) \approx D_r \log \sum_j w_j f(Y|z_j)$. The expectation of the latter may be approximated by another quadrature

$$\begin{aligned} k_r &= \sum_y \left(D_r \log L(\mu, \tau^2|y)\right) \int f(y|z)\phi(z)dz \\ &\approx \sum_y \left(D_r \log L(\mu, \tau^2|y)\right) \sum_k w_k f(y|z_k) \\ &\approx \sum_y \left(D_r \log \sum_j w_j f(y|z_j)\right) \sum_k w_k f(y|z_k) \end{aligned}$$

The outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson model. The Poisson model requires a truncation rule, an additional tuning parameter discussed further below. The general tensor component needed for the computation of the Bartlett correction factor is then approximated as

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|z_j)\right) \sum_k w_k f(y|z_k)\right)$$

Carrying the differentiations ((ref above display)) through yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$, say, $F_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k} \frac{d^l}{d(\tau^2)^l} f(y|z_j)$. The 35 tensor components ((k)), in terms of the F_{kl}, are the same for any model based on a likelihood of the form ((ref)), and may be pre-computed. For example $k_{\mu,\mu} = \sum_y (F_{20}(y) -$

$(F_{10}(y)^2/F_{00}(y)))$, and $k_{\mu,\tau^2}^{(\tau^2)} = \sum_y (F_{01}(y)^2 F_{10}(y)/F_{00}(y)^2 - (F_{02}(y)F_{10}(y) + F_{01}(y)F_{11}(y))/F_{00}(y) + F_{12}(y))$.

The derivatives $F_{jk}(y)$ of the model-specific conditional PMFs that appear in the 35 tensor components number 15: $F_{jk}(y)$ for $0 \leq j + k \leq 4$. The form of these derivatives may be derived independently of data but their specific values in general depend on parameters of the conditional PMF ((see above)), and must typically be computed for each study of a meta-analysis. The latter computations may be carried out in parallel when the meta-analysis consists of stochatically independent studies, such as we consider here. Furthermore, since the conditional PMFs only depend on the model parameters (μ, τ^2) through the PMF's distributional parameters, the derivatives $F_{jk}(y), 0 \leq j + k \leq 4$, may be re-written as linear combinations [[linear for binomial and hyperg also??]] of $F_{00}(y) = \sum_j w_j f(y|\theta_j)$ using standard recurrences ((cite johnson and kotz univariate discrete distributions)). The advantage of this form is that the terms F_{00} involve only the PMF itself, for which optimized and numerically stable routines are available. For example, two applications of the Poisson PMF recurrence $\frac{d}{d\lambda} f_\lambda(y \mid Z) = f_\lambda(y-1 \mid Z) - f_\lambda(y \mid Z), y \geq 1$, yield, for model ((ref poisson/incidence rate model)),

$$\begin{aligned} F_{11}(y) &= \sum_j w_j \frac{d^2}{d\mu d\tau^2} f_\lambda(y \mid z_j) \\ &= \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y-2) + (\frac{d^2\lambda}{d\mu d\tau^2} - 2\frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2}) F_{00}(y-1) + \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} F_{00}(y), \qquad y \geq 2 \end{aligned}$$

further implementation details. two hyperparameters for poisson. speed issues with hyperg. also the grid size for profile curve–this is present already with the LR statistic.

4 Simulation

we examine by simulation the rate at which the LR and Bartlett-corrected LR statistics converge to the reference chi-squared distribution under the models ((ref)). The purpose is three-fold: 1. to investigate whether the quadrature approximations used by the proposed method are sufficient, 2. to investigate whether the bartlett correction is effective as applied to this type of lattice data, as there are few theoretical results here, and 3. to assess the robustness the of the proposed method to deviations from the assumed Gaussian model for the random effect.[[add tau2 boundary issue here]]

to make the simulation results relevant, a meta-analysis of we data parameters are sampled from stijnen tables ((data analysis)) so as to resemble real data. we also examine robustness of the gaussianity assumption. since the leading term in CLT-type approximations is based on the skew, we examine how the rate varies as the skew of the underlying RV is varied. we use the sinh-asinch model ... to obtain a family of RVs based on the Gaussian with the skew perturbed. examine therate approximated using log-log.

5 Data analysis

We revisit the data analysis of ((stijnen2010)).

4a. proportions
4b. incidence